

QuickVoice Master Admin Configuration Guide

Complete Administrator Walkthrough for Production Setup, LiveKit Cloud, Deepgram STT/TTS, Inbound/Outbound Webhooks, Agent API Configuration, and System Health

Product Version:	QuickVoice Enterprise v2.5
Classification:	System Administration / DevOps
Primary Integrations:	LiveKit Cloud, Deepgram, Twilio/Telnyx, LLMs
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1. Environment Configuration & Master Secrets

1.1 Architectural Overview of Services

The QuickVoice platform requires synchronized configuration across three distinct service domains:

- 1. **API Server** (`apps/server/.env`): Manages authentication, database, billing, webhook routing, and telephony signaling.
- 2. **Console Dashboard** (`apps/console/.env`): Client-side Next.js web application.
- 3. **AI Voice Worker** (`apps/ai/.env`): Python 3.11 service connecting directly to WebRTC media rooms, speech models, and real-time inference.

1.2 Critical Security Secrets (Must Configure First)

Variable Name	Location	Requirement & Purpose	Generation Command
<code>SECRET_ENCRYPTION_KEY</code>	<code>apps/server/.env</code>	64 Hex Characters (32 Bytes). Used to encrypt third-party tool credentials, API tokens, and webhook secrets at rest via AES-256-GCM.	<code>openssl rand -hex 32</code>
<code>INTERNAL_API_KEY</code>	<code>apps/server/.env</code> & <code>apps/ai/.env</code>	High-Entropy Shared Token. Secures internal inter-service communication between Node.js API and the Python AI worker.	<code>openssl rand -base64 24</code>
<code>BETTER_AUTH_SECRET</code>	<code>apps/server/.env</code>	32+ Character Secret. Signs JWT session tokens and user authentication cookies.	<code>openssl rand -base64 32</code>

Critical Production Rule

`INTERNAL_API_KEY` must match identically between `apps/server/.env` and `apps/ai/.env`. If mismatched, the AI worker will be rejected with HTTP 401 when verifying billing tokens or reporting call metrics.

2. Connecting LiveKit Cloud API & Telephony

2.1 Why LiveKit Cloud?

LiveKit provides the carrier-grade WebRTC media plane and SIP gateway. While LiveKit can be self-hosted, **LiveKit Cloud** provides a globally distributed edge mesh with zero UDP port-forwarding issues and 5,000 free minutes/month.

2.2 Step-by-Step LiveKit Cloud Provisioning

STEP 1 Create a Project in LiveKit Cloud

Sign up at <https://cloud.livekit.io> \$ ightarrow\$ Create a new project named `quickvoice-prod`.

STEP 2 Generate API Credentials

In the LiveKit Cloud dashboard, go to **Project Settings** \$ ightarrow\$ **Keys** \$ ightarrow\$ Click **Generate Key**.
Copy the following three values into your `apps/server/.env` and `apps/ai/.env`:

```
LIVEKIT_URL=wss://quickvoice-prod-xxxx.livekit.cloud
LIVEKIT_API_KEY=APIxxxxxxxxxxxxxxxx
LIVEKIT_API_SECRET=xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
LIVEKIT_AGENT_NAME=quickvoice-voice-agent
```

STEP 3 Create LiveKit SIP Inbound Trunk

1. In LiveKit Cloud, navigate to **SIP** \$ ightarrow\$ **Inbound Trunks** \$ ightarrow\$ Click **Create Inbound Trunk**.
2. Set Name to: `quickvoice-inbound`.
3. Set Allowed Numbers or leave open to accept all incoming carrier calls.
4. Copy the resulting Trunk ID (e.g. `ST_inbound_xxxxxx`) into `apps/server/.env`:

```
LIVEKIT_SIP_INBOUND_TRUNK_ID=ST_inbound_xxxxxx
```

STEP 4 Create LiveKit SIP Outbound Trunk (Twilio / Telnyx)

1. Navigate to **SIP** \$ ightarrow\$ **Outbound Trunks** \$ ightarrow\$ Click **Create Outbound Trunk**.
2. Address: `your-carrier-trunk.pstn.twilio.com` (or Telnyx SIP endpoint).
3. Auth Username & Password: Set from your Twilio Elastic SIP Trunk credentials.
4. Copy the Trunk ID into `apps/server/.env`:

```
LIVEKIT_SIP_OUTBOUND_TRUNK_TWILIO_ID=ST_outbound_twilio_xxxx
LIVEKIT_SIP_OUTBOUND_TRUNK_TELNYX_ID=ST_outbound_telnyx_xxxx
```

2.3 LiveKit Dispatch Rule (Connecting Phone to AI Worker)

In LiveKit Cloud SIP settings, create a **Dispatch Rule**:

- **Trigger:** Any incoming SIP call on `quickvoice-inbound`.
- **Action:** Create Room with prefix `voice-` and dispatch the registered agent `quickvoice-voice-agent`.

3. Connecting Deepgram STT & TTS Pipeline

3.1 Generating Deepgram API Credentials

- 1. Register at <https://deepgram.com>.
- 2. Navigate to **API Keys** → Click **Create Key**. Select permissions: *Member* or *Admin*.
- 3. Add the API key to `apps/ai/.env`:

```
DEEPGRAM_API_KEY=dg_sec_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

3.2 Deepgram Speech-to-Text (STT) Options & Latency Tuning

In QuickVoice, STT operates via WebSocket streaming using interim chunked results:

Parameter	Config Value	Recommendation
<code>model</code>	<code>nova-3</code> (Recommended) or <code>nova-2</code>	Nova-3 delivers lowest word error rate (WER) and sub-180ms streaming latency.
<code>language</code>	<code>en</code> (or <code>es</code> , <code>fr</code> , <code>de</code> , <code>hi</code>)	Select primary agent language or pass multi-language tag.
<code>smart_format</code>	<code>true</code>	Automatically formats dates, currencies, phone numbers, and punctuation.
<code>interim_results</code>	<code>true</code>	Streams interim transcripts to Silero VAD for instant user interruption detection.
<code>endpointing</code>	<code>250 ms</code>	Silence duration required before committing the caller turn to the LLM.

3.3 Deepgram Aura Text-to-Speech (TTS) Voice Catalog

QuickVoice natively supports Deepgram Aura-2 high-speed streaming voices:

Voice ID	Gender	Persona Tone	Ideal Use Case
<code>aura-2-asteria-en</code>	Female	Warm, confident, conversational	Inbound Receptionist & Customer Support
<code>aura-2-hera-en</code>	Female	Authoritative, professional	Medical, Legal & Financial Intake
<code>aura-2-zeus-en</code>	Male	Deep, friendly, trustworthy	B2B Sales Qualification & Outbound Dialing
<code>aura-2-luna-en</code>	Female	Gentle, empathetic	Healthcare Appointment Reminders

Alternative TTS Providers

QuickVoice also supports **ElevenLabs** (`ELEVENLABS_API_KEY` , model: `eleven_flash_v2_5`) and **Cartesia** for ultra-expressive emotional voices.

4. Configuring Inbound & Outbound Webhooks

4.1 Agent Initiation Webhook (Pre-Call Context Injection)

Before an inbound or outbound call begins, QuickVoice can query your CRM or database to fetch dynamic variables (caller name, past order history, balance, account tier) to personalize the conversation.

Configuration in Agent Settings:

```
{
  "initiation_webhook": {
    "url": "https://api.yourcompany.com/crm/caller-lookup",
    "method": "POST",
    "headers": {
      "Authorization": "Bearer YOUR_CRM_WEBHOOK_SECRET",
      "Content-Type": "application/json"
    },
    "timeout_secs": 4
  }
}
```

Payload Sent to Your Endpoint:

```
{
  "event": "call.initiation",
  "callId": "call_987654321",
  "callerNumber": "+14155550199",
  "agentId": "e1687102-91d1-47bb-a7d9-d43e89a55298",
  "direction": "inbound",
  "timestamp": "2026-09-04T06:00:00.000Z"
}
```

Expected Return Payload (Injected into Agent Variables):

```
{
  "variables": {
    "caller_name": "Alexander Vance",
    "account_tier": "VIP Gold",
    "last_order_status": "Out for delivery"
  }
}
```

The agent can immediately speak: *"Hi Alexander, welcome back to QuickVoice VIP support! Are you calling about your order that is out for delivery today?"*

4.2 Post-Call Webhook (CRM Sync, Summary & Analytics)

Triggered immediately after a call terminates. Sends full conversation metadata, structured data slots, and audio recording URLs.

Payload Dispatched by QuickVoice:

```
{
  "event": "call.completed",
  "callId": "call_987654321",
  "agentId": "e1687102-91d1-47bb-a7d9-d43e89a55298",
  "durationSeconds": 142,
  "status": "COMPLETED",
  "callerNumber": "+14155550199",
  "recordingUrl": "https://s3.amazonaws.com/quickvoice-recordings/call_987654321.mp3",
  "transcript": [
    { "role": "agent", "message": "Hi, thanks for calling! How can I assist you today?" },
    { "role": "caller", "message": "I'd like to book an appointment for tomorrow at 2 PM." },
    { "role": "agent", "message": "You're booked for tomorrow at 2 PM with Dr. Vance!" }
  ],
  "extractedData": {
    "appointment_date": "2026-09-05",
    "appointment_time": "14:00",
    "user_email": "alex@example.com"
  },
  "sentiment": "POSITIVE",
  "escalatedToHuman": false,
  "telephonyCostMicros": 124000
}
```

5. Creating & Configuring Voice Agents in Console UI & API

5.1 Step-by-Step Agent Creation through the Console UI

STEP 1 Create Agent Entity in Console

In the main sidebar, navigate to **Agents** (<http://localhost:3005/agents>) → Click the + **Create Agent** button in the top right. Enter a friendly name (e.g., "Apex Dental Receptionist") and click **Create**.

STEP 2 Open the "Voice & Models" Configuration Tab

From the agent overview screen, click the **Voice** tab in the navigation bar. This opens the dedicated model and speech engine configuration panel.

5.2 Configuring LLM, STT, and TTS in the Console UI

QuickVoice provides decoupled dropdown selectors allowing you to mix-and-match any combination of intelligence, transcription, and speech synthesis:

1 Agent Language Selector (`agent_language`)

Select the conversational language (e.g., **English (en)**, **Spanish (es)**, **French (fr)**, **German (de)**, **Hindi (hi)**). Changing the language dynamically filters the available STT and TTS models below to only those supporting that locale.

2 LLM Model Selector (`llmModel`)

Select the reasoning engine that generates the conversation turns:

- **Claude 3.5 Haiku (Amazon Bedrock / Anthropic)**: Recommended default. Delivers sub-200ms first-token latency with superior prompt adherence.
- **Claude 3.5 Sonnet**: Used for highly complex reasoning, multi-turn diagnostics, or complex objection handling.
- **Amazon Nova Micro / Lite**: Cost-effective options for straightforward deterministic IVR workflows.
- **OpenAI GPT-4o-mini**: Ultra-fast general intelligence for support intake and lead qualification.

3 Speech-to-Text (STT) Model Selector (`sttModel`)

Select the real-time transcription engine that listens to the caller:

- **Deepgram Nova-3 (`deepgram/nova-3`)**: Lowest word error rate (WER) and streaming latency (<180ms). Recommended for English and Spanish.
- **Deepgram Nova-2 (`deepgram/nova-2`)**: Battle-tested streaming STT with multi-lingual support.
- **Sarvam Saaras (`sarvam/saaras:v3`)**: Optimized for Indian accents, vernacular dialects, and code-switching (Hinglish).

4 Text-to-Speech (TTS) Model & Voice Selection (`ttsModel` & `voiceId`)

Choose the synthesis engine and specific human persona voice:

- **Deepgram Aura-2** (`deepgram/aura-2`): Instant streaming audio. Choose from *Asteria* (energetic/clear), *Apollo* (confident), *Hera* (warm/professional), *Zeus* (deep baritone), or *Luna* (gentle).
- **ElevenLabs Flash v2.5** (`elevenlabs/eleven_flash_v2_5`): Ultra-expressive emotional nuance for high-touch customer VIP lines.
- **Rime** (`rime/rime-arcana`): Studio-grade natural conversational inflection.

5 Audition Voice with the Interactive Preview Panel

The right-hand side of the **Voice Tab** displays the **Voice Profile Panel**. Click the **Play Sample Audio** button to audition the voice's pitch, cadence, and tone before saving. You can also view its style tags (e.g., "*Clear*", "*Confident*", "*Energetic*") and recommended use cases.

6 Click "Save Voice Configuration"

Click the **Save Voice Configuration** button in the top right. The console dispatches an authenticated `PUT /api/v1/agents/{agentId}/configure` request, updating the live agent configuration immediately.

5.3 How Administrators Add Custom Models to the UI Catalog

If your organization wants to expose custom fine-tuned LLMs (e.g. self-hosted vLLM or Ollama), new ElevenLabs voice clones, or private STT models inside the Console UI dropdowns:

1. **Catalog File:** Edit `apps/console/src/lib/data/voices.ts` (or provide a custom JSON file via `VOICE_CATALOG_PATH` in `apps/ai/.env`).
2. **Add New LLM Option:**

```
export const LLM_MODELS: ModelOption[] = [
  ...LLM_MODELS,
  {
    id: "openai/gpt-4o-mini",
    label: "OpenAI GPT-4o Mini",
    provider: "OpenAI"
  }
];
```

3. **Add Custom Voice ID:**

```
{
  id: "my-custom-cloned-voice",
  name: "Dr. Sarah Cloned Voice",
  provider: "ElevenLabs",
  gender: "feminine",
  locale: "en-US",
  accent: "American",
  languages: ["en"],
  ttsModels: ["elevenlabs/eleven_flash_v2_5"],
  styles: ["Professional", "Medical"],
  useCases: ["Dental Clinics", "Healthcare"]
}
```

4. The new model and voice will automatically populate inside the Console UI dropdowns for all users in the organization!

5.4 Programmatic Agent Creation & Configuration via REST API

For automated deployments and external integrations, administrators can execute the exact same actions via API:

Step 1: Create the Agent Entity

```
POST /api/v1/agents
Content-Type: application/json
Origin: http://localhost:3005
Cookie: better-auth.session_token=YOUR_SESSION_TOKEN

{
  "name": "Concierge Receptionist",
  "isActive": true
}
```

Step 2: Full Agent Configuration (LLM, STT, TTS, Prompts, Webhooks)

```
PUT /api/v1/agents/{agentId}/configure
Content-Type: application/json
Origin: http://localhost:3005
Cookie: better-auth.session_token=YOUR_SESSION_TOKEN

{
  "agent_language": "en",
  "firstMessage": "Hi, thanks for calling! How can I help you today?",
  "systemPrompt": "You are a concise concierge voice receptionist. Keep responses under 20 words. Verify details",
  "llmModel": "bedrock/us.anthropic.claude-haiku-4-5-20251001-v1:0",
  "sttModel": "deepgram/nova-3",
  "ttsModel": "deepgram/aura-2",
  "voiceId": "aura-2-asteria-en",
  "store_call_audio": true,
  "zero_pii_retention": false,
  "conversation_retention_days": 30,
  "ivr_navigation_enabled": true,
  "timezone": "America/New_York",
  "data_needed": [
    { "id": "email", "type": "string", "name": "Email", "description": "Caller email address" }
  ]
}
```

Step 3: Attach an Integration Tool (e.g. Cal.com or SMS):

```
POST /api/v1/tools/{toolId}/attach/{agentId}
```

Step 4: Assign an Inbound Telephony Number:

```
POST /api/v1/phone-numbers/{phoneNumberId}/assign
Content-Type: application/json

{
  "agentId": "{agentId}"
}
```

6. Verification, Health Checks & Troubleshooting

6.1 System Health & Readiness Checks

Before launching live campaigns, run the following automated diagnostics from your terminal:

Diagnostic Check	Command	Expected Output
API Readiness Check	<code>curl http://localhost:5000/api/v1/system/readiness</code>	<code>{"status":"ready","database":true,"redis":true}</code>
Swagger Documentation	<code>curl -I http://localhost:5000/api/v1/docs</code>	<code>HTTP/1.1 200 OK</code>
Console Web UI	<code>curl -I http://localhost:3005/login</code>	<code>HTTP/1.1 200 OK</code>
AI Worker Port	<code>curl http://localhost:5555/health</code>	<code>{"status":"healthy","livekit":true}</code>

6.2 Administrator Troubleshooting Matrix

Error / Symptom	Root Cause	Remediation Action
<code>MISSING_OR_NULL_ORIGIN</code>	Client API call missing <code>Origin</code> header in Better Auth requests.	Include <code>Origin: http://localhost:3005</code> in curl requests or configure <code>CORS_ORIGINS</code> in <code>apps/server/.env</code> .
<code>listen EADDRINUSE :3000</code>	Port 3000 occupied by local container (e.g. Open WebUI / Grafana).	Console has been configured to use Port 3005 (<code>next dev -p 3005</code>).
<code>LiveKit Room Not Joining</code>	Mismatched <code>LIVEKIT_URL</code> or expired <code>LIVEKIT_API_SECRET</code> .	Regenerate key in LiveKit Cloud dashboard; update both <code>apps/server/.env</code> and <code>apps/ai/.env</code> .
<code>STT Not Transcribing Audio</code>	Invalid or depleted Deepgram API balance.	Check Deepgram console credit balance; ensure <code>DEEPGRAM_API_KEY</code> is active.
<code>Negative Balance / Call Blocked</code>	Wallet reservation failed due to empty balance.	Credit organization wallet via Stripe or seed balance using <code>prisma.billingAccount.update()</code> .

Need Production Assistance?

For enterprise support, custom telephony SIP gateway integrations, and managed cloud deployments, consult your repository: <https://github.com/mithuncdev/Advanced-quick-voice.git>